



Grid-connected hydro–solar hybrid systems for low-carbon energy transition: a techno-economic assessment

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Abstract

To support the global transition from fossil fuel to cleaner energy sources, this study conducts the techno-economic analyses of four hybrid energy systems: Grid/photovoltaic (PV)/Hydro, PV/Hydro, Grid/Hydro, and Grid/PV for sustainable rural electrification in Nigeria. While hybrid energy systems using solar PV, wind, and diesel have attracted credible attention, this work highlights the untapped potential of run-of-river hydro-power, a carbon-free resource with widespread networks across the globe. The analyses using the HOMER pro software tool incorporated the incessant utility grid interruption characteristics of developing countries. Results reveal that the hybrid Grid/PV/Hydro system is optimal in meeting the community's electricity demand of 2,822.20 kWh/day at the lowest net present cost of \$9,160,308.00 and levelized cost of energy of \$0.13/kWh. The micro-hydro power plants, solar PV, and grid, respectively, contribute 80.90%, 12.4%, and 6.76% of the total electricity generation by the system. While the Grid/PV system is emission-free, the CO₂ emission of the optimal system is 55.46% and 0.09% of Grid/Hydro and Grid/PV systems, respectively. Sensitivity analyses indicate that the river flow rates, discount rates, inflation rates, and load demand significantly impact the optimal system's economics, in contrast to the capacities of the components, electricity generation, and operating strategy. This study is a worthwhile framework for policy makers and renewable energy developers for mitigating the twin issues of energy poverty and carbon emissions plaguing the world, particularly the developing economies.

Keywords Climate change · Greenhouse gas emission · HOMER · Hybrid renewable energy systems · Hydropower · Run-of-river

Nomenclatures

RE	Renewable energy
RF	Renewable fraction
GHG	Greenhouse gas
HRES	Hybrid renewable energy systems
LCOE	Levelised cost of energy

Extended author information available on the last page of the article

NPC	Net present cost
WT	Wind turbine
DG	Diesel generator
PV	Photovoltaic
MHT	Micro-hydro turbines
BG	Biogas generator
HOMER	Hybrid Optimization Model for Electric Renewables
DC	Direct current
CRF	Capital recovery factor
AC	Alternating current
O&M	Operation and maintenance
LF	Load following
BESS	Battery energy storage systems
CC	Cycle charging
SoC	State of charge
CO ₂	Carbon dioxide
CO	Carbon monoxide
SO ₂	Sulphur dioxide
UH	Unburned hydrocarbons
PM	Particulate matter

1 Introduction

The Paris Agreement, adopted in 2015, aims to address climate change by confining the rise in the mean global temperature to well below 2 °C and working toward a 1.5 °C limit. To cap the temperature rise in the Agreements, carbon dioxide (CO₂) emissions must be kept below 1.1 Ttons by ensuring that about 85%, 50%, and 34% of the global coal, natural gas, and oil reserves, respectively, are unexploited before 2050 (Shen et al., 2018). This is a stringent task since the main contributor to climate change is the greenhouse gas (GHG) emissions from the massive utilization of fossil fuels (coal, oil, and natural gas) to mitigate the global energy deficit (Dodo et al., 2024).

Electricity, being a nucleus of socio-economic activities, cannot be relegated. Therefore, it is concerning that, despite being an instrument for the general well-being and growth of the populace, about 759 million global population do not currently have access to electricity, while in Sub-Saharan Africa, about 60% of the population lives in areas with poor electricity access, and 58% do not have it at all (Saddari et al., 2025). Furthermore, the gap between global energy demand and supply has continued to widen. The next ten years are envisaged to witness a surge in the global energy demand by over 30% (Elmasides et al., 2025). Consequently, the desire to address these energy demand–supply gaps has stimulated the energy planners and the academic community to intensify efforts and novel interests in the deployment of renewable energy (RE) technologies (Elmasides et al., 2025; Hussain Alhamami et al., 2024). This is crucial not just for tackling the global energy crisis but also for minimizing the impacts of climate change, such as increasing temperatures and severe weather conditions.

Recent years have seen notable advancements in RE technologies to the point that they currently contribute significantly to energy generation. In Europe, for instance, the share of RE output increased from 14 to 29% annually between 2005 and 2017 (Frate et al., 2021). Meanwhile, the deployment of RE technologies needs to progress significantly in developing countries, particularly sub-Saharan Africa, because their electricity industries, despite grappling with insufficient generation capacities, still mostly utilize coal and natural gas (Husein et al., 2024). Therefore, a variety of contextual factors must be taken into account to fully understand the complexities of electricity crises. Notably, every country faces different obstacles that call for flexible approaches to guarantee a steady supply of electricity. In Nigeria, for instance, the national grid, which provides 51.2% of the country's electricity, finds it difficult to keep up with the growing electricity demand (Ezekwem et al., 2024; Odetoeye et al., 2023), while the application of the abundant renewable resources to complement the existing power infrastructure is at its lowest ebb. Out of the mean 4,000 MW available for over 200 million population, the RE share, primarily hydro-power, is 14%, while the remaining 86% comes from fossil fuels (coal and natural gas) (Dodo & Ashigwuike, 2023). This is a country whose aggregate wind, solar, hydro, and biomass energy potentials are estimated as 1,182.4 GWh per day (Akhator et al., 2019).

Similar to wind and solar energy systems, the advancements in technology have made hydroelectricity appealing. It has relatively low operating and maintenance costs, the ability to withstand peak power demand, and nearly zero carbon emissions (Dodo et al., 2021). Under the charter of the Paris Agreement and Sustainable Development Goals, hydropower schemes guarantee that nations are shielded from the worst effects of climate change, such as drought and flooding. Meanwhile, destruction to wildlife habitations and migration paths, alterations to the regional precipitation scheme, regulated water discharge to the river downstream of the dam, compelled relocation of communities or settlements beneath the reservoir, and other detrimental impacts are some of the most often cited downsides of hydro-power plants. Notwithstanding all these drawbacks, hydro-power generation offers the least expensive method of producing electricity compared to fossil fuels, solar, onshore wind, offshore wind, geothermal, and biomass, respectively (Bayazit, 2021).

The technological architecture of hydropower facilities can be either run-of-river (weir or barrage with little or no pondage) or storage (dam and reservoir-based). The production of electricity through run-of-river plants is an emerging option for storage-based hydropower plants (Ibrahim et al., 2019). In a run-of-river, a portion of stream flow is redirected through penstocks to a powerhouse for electricity generation, after which it is redirected back into the stream. Besides its substantially shorter construction time and lower overall costs compared to reservoir-based hydropower plants, the main advantage of this technology is that it does not require a large reservoir for operation (Bawan et al., 2025). Such simplicity of the run-of-river scheme makes it attractive, but it is also its main weakness. Since the river is not dammed, the power plant is solely dependent on the river's water flow. Hence, the quantity of electricity is affected by the seasonal flow. When the river is flooded, most of the available water must be permitted to run past the diversion system unutilized, whereas, during a drought, the plant will be unable to produce any electricity. However, there are other important benefits to this kind of project than cost, especially considering how little environmental disturbance it creates.

The drawbacks of run-of-rivers and the stochastic nature of other RE sources, such as solar and wind, can hamper continuous electricity generation and also damage the grid

components (motors, transformers, and generators). Hence, it is essential to combine these diverse energy sources with robust energy storage facilities for enhanced power flow, grid stability, continuous power generation, and prolonging the lifespan of the power components (Dodo et al., 2024). Such aggregation of diverse energy sources for complementary operation constitutes hybrid renewable energy systems, which can be operated in grid-connected or stand-alone modes. The combinations of distinct energy sources make the efficient planning and operation of HRES highly challenging. Thus, a lack of proper planning may lead to high installation costs, over/under-sizing of the components of the system, or even a capacity shortage. Hence, it is crucial to meticulously investigate the viability of every HRES. This will help to identify an optimal option that will guarantee the sustainability and reliability of electricity generation with minimal costs and the least environmental impact.

1.1 Review of similar studies

Within the academic community, studies on HRES have continued to garner interest ranging from reducing the energy demand–supply gaps (Ali et al., 2025a, b; Demir, 2024; Yahya et al., 2024), efficient energy management and dispatch strategies (Adetoro et al., 2023; Agou-ndedemba et al., 2025; Elmasides et al., 2025; Vu et al., 2019), identifying optimum configuration of power components (Allouhi, 2024; Al-Omari et al., 2025; Dodo et al., 2022; Nemouchi et al., 2024), and selection of optimal energy storage technologies (Akinte et al., 2023; Dodo et al., 2024). For instance, Allouhi (2024) investigated an off-grid solar PV-WT-FC-micro-hydro system to supply Ouenskra, a rural location in Morocco, with clean and reliable electricity. Improving energy availability and encouraging sustainable development in the area are the primary goals. To establish a green and sustainable campus, Demir (2024) focused on providing uninterrupted and clean electricity to Aksaray University using off-grid PV-WT-FC systems. Although the optimal system presented a high leveled electricity cost of 0.327\$/kWh, it succeeded in providing clean fuel to facilitate the staff and students' transportation using eco-friendly hydrogen fuel cells. The competitive performance of genetic algorithms (GA), whale optimization algorithm (WOA), ant colony optimization (ACO), and particle swarm optimization (PSO) for sizing HRES was investigated by Adetoro et al. (2023). The algorithms were tested for two distinct districts of Darnah and Alkhums in Libya. Darnah had a cost of energy (COE) of \$1.875 via WOA, while the Alkhums district had the lowest COE of \$1.86, utilizing PSO. Bawan et al. (2025) examined five distinct energy systems, such as DG, DG/Hydro, DG/PV, DG/Hydro/PV, and PV/Hydro, to identify the optimal architecture that will satisfy the electricity requirements of a district in Indonesia. Elmasides et al. (2025) developed a comprehensive power management strategy for a PV-dominated HRES. The DG and PEM fuel cells and alkaline electrolyzers were integrated into the system for complementary operation. The battery's state of charge (SoC) was set to fluctuate around 20 to 95% of its nominal capacity, forming the basis for the operating strategy. Al-Omari et al. (2025) investigated an optimum configuration of WT/PV/DG/battery energy systems to energise a water pumping system in a remote desert of Al-Mudawwara village/Ma'an governorate in Jordan. Using the HOMER software tool, the scenario with DG/WT/storage batteries presents an optimal solution. The Seagull Optimization Algorithm (SOA) was utilized by Nemouchi et al. (2024) to size HRES comprising PV/WT/DG for irrigation systems in Algeria. The NPC, LCOE, loss of power supply probability (LPSP), and loss of water supply probability of the optimal system, respectively,

are \$14,140, 0.292 \$/kWh, 24%, and 41.2%. Ali et al., (2025a, b) proposed a microgrid to address Bangladesh's significant and pressing problem of frequent power outages in a rural elementary school. The HOMER Pro was used to analyze four distinct system configurations consisting of PV/WT/BESS and the grid. The most reliable, cost-effective, and environmentally benign architecture is PV-WT-BESS, consisting of two strings of 12.8 V, 100 Ah BESS, six units of 1 kW WT, and 13.3 kW solar PV. The COE, NPC, and RF, respectively, are \$0.0363/kWh, \$10,981.38, and 81%. The grid emission characteristics were excluded from the analysis. Agoundedemba et al. (2025) used genetic algorithms GA and model predictive control to size and optimize the hybrid PV/Wind/FC/Battery system. The primary goal was to minimize the NPC, LPSP, and COE. The optimal system produces 17.29 GWh of energy annually at \$0.19/kWh and an LPSP of 0%. A HRES that includes PV, WT, and DG for a modest domestic load and a large load, such as a hostel, was examined by Rajbongshi and Rajkumari (2025). The HOMER software tool was used to determine off-grid and grid-connected systems that have the lowest COE. With a COE of ₹10.66/kWh for the hostel and ₹12.48/kWh for the domestic load, the hybrid system that consists of PV, DG, battery, and converter provides the best off-grid system configuration. Al-Rawashdeh et al. (2023) proposed grid-connected PV/WT//DG HRES to lower the COE of a hotel in Petra, Jordan, through the examination of various configurations of sustainable energy resources. Four scenarios were considered: PV/DG/Grid, WT/DG/Grid, PV/WT/DG/Grid, and PV/WT/DG/Grid. The best configuration in terms of economy and technology is Grid/PV/DG.

1.2 Contributions of the study

Hybrid solar, wind, and diesel energy systems have been foremost in the global effort to address the energy deficit and climate change, as depicted through the review of similar studies. In contrast, the integration of hydro-power plants harnessing run-of-river into the energy mix, despite the extensive networks of rivers that navigate the globe, has not attracted novel interest. Furthermore, the focus on HRES research is substantially on off-grid systems, while the few studies that concentrated on grid-connected systems overlooked grid failure circumstances. Such studies cannot reflect the realities of developing countries, such as Nigeria, where the grid interruptions are frequent, with durations lasting for hours or days.

To address the aforementioned literature gaps, this study aimed to investigate the techno-economic and environmental viability of hybrid run-of-river and solar energy systems connected to an erratic utility grid. Two noteworthy features of this study are the exclusion of traditional diesel generators that are frequently utilized in similar circumstances and the consideration of frequent grid interruptions to reflect the realities of developing economies. A series of simulations and optimizations of four distinct architectures of energy systems (Grid/PV/Hydro, PV/Hydro, Grid/Hydro, and Grid/PV) was performed using a micropower optimization model with the objective function of minimizing the net present cost (NPC). The key components of the technical analysis include the capacities of the various components, energy storage profiles of the battery bank, and energy production of the systems, while the economic analysis dwells on the system NPC, salvage costs, LCOE, capital costs, replacement costs, and operating costs. The environmental analysis provides insight into the emissions associated with the proposed energy systems. Since hydro and solar resources are carbon-free, the implementation of this study, besides improving electricity access, will help

to curb GHG emissions, which are central to the attainment of the United Nations' sustainable development goals 3 and 7.

2 Materials and methods

In this study, techno-economic feasibility analyses of hybrid energy systems harnessing run-of-river and solar resources for a community suffering from intermittent grid failures in Nigeria were performed. To fully explore the optimal electrification schemes for the community, four distinct scenarios, namely, Grid/PV/Hydro, Grid/Hydro, PV/Hydro, and Grid/PV systems, were considered. A hybrid optimization of multiple energy resources (HOMER Pro) was utilized to analyse a variety of design options. HOMER Pro was chosen for its accuracy, simplified optimization, efficient analytics, customizable design, and insightful interfacing (He et al., 2023; Malik et al., 2022). HOMER Pro presents the various aspects of the energy systems in tables, charts, and graphs, allowing the user to compare various configurations and guaranteeing more control for proper planning. The schematics of the investigated hybrid energy systems are shown in Fig. 1. The converter synchronizes DC to AC and vice versa to ensure a free flow of electricity between the DC and AC buses while the battery bank serves as a backup electricity storage that could come on-board in the event of grid failures or insufficient generation from the PV panels and micro-hydro turbines (MHT). The grid electricity flow is unidirectional, allowing for purchase without selling back the

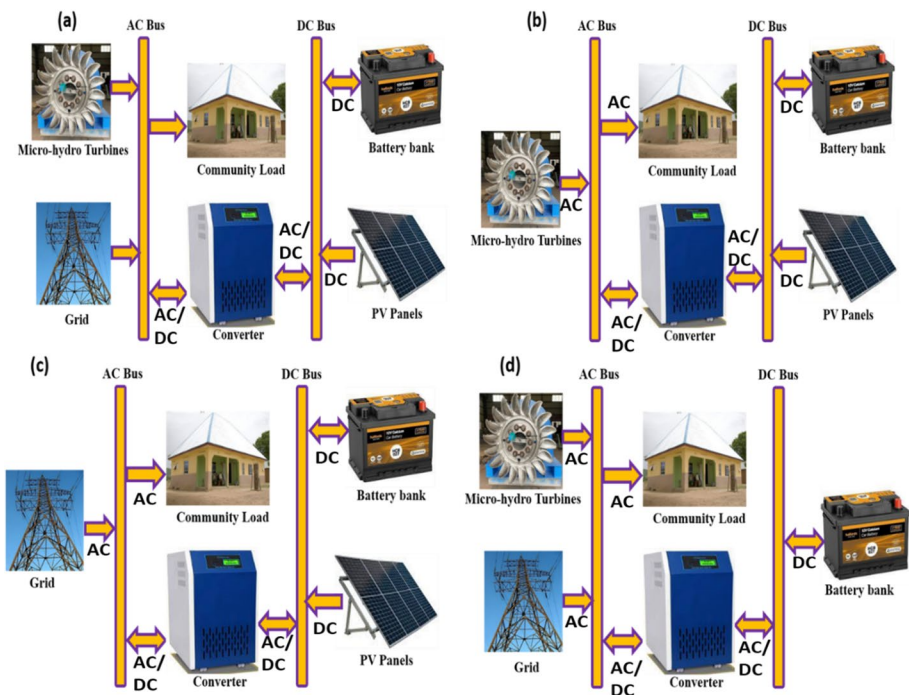


Fig. 1 Schematics of the hybrid energy systems (a) Grid/PV/MHT, (b) PV/MHT, (c) Grid/PV, (d) Grid/MHT

excess electricity that could be generated by the system. This is due lack of enforcement of net metering by the Nigerian Electricity Regulatory Commission.

2.1 Description of the study location- Giri community

Giri is a rural area in the Gwagwalada Area Council of the Federal Capital Territory of Nigeria. Situated on latitude 8.9508° N and longitude 7.0767° E, the community is characterized by a mean temperature of 27.3 °C, annual rainfall of about 1,460 mm, and a run-of-river of 1,330 mm in a catchment area of over 30 km² (Ozigis et al., 2020). Although the inhabitants are predominantly farmers, the community provides an off-campus residence for an array of University of Abuja staff and students. The community's electricity supply from the grid is epileptic, and due to the fuel subsidy removal by the Nigerian government in May 2023, the exorbitant costs of running diesel and gasoline generators have continued to be unbearable to the people. This not only jeopardizes the sustainability of businesses but also impedes the community's larger attempts at socio-economic growth.

2.2 Load assessment of the community

Electricity demand assessment is the first step in planning every power system to ensure a consistent and sustainable supply of electricity. The electricity consumption was evaluated in terms of the quantity of appliances, the wattage, and the estimated duration of energy usage per appliance, characteristic of rural communities in Nigeria (Dodo et al., 2022). The total electricity demand ($P_T^{(i)}$) by the community at each hour (i) was computed using eqs. (1) and (2).

$$P_T^{(i)} = \sum_{k=1}^M n_k \times P_k \quad (1)$$

where M is the cumulative quantity of appliances in use at an hour (i), n_k is the quantity of appliance k in use, and P_k is the electricity consumption of appliance. Therefore, the daily cumulative electricity demand of the community was estimated using Eq. (2).

$$E_T = \sum_{i=1}^{24} P_T^{(i)} \quad (2)$$

A total of 250 households, two eateries, 2 barbershops, one primary school, 2 hair salons, one clinic, two worship centres, and a community water supply scheme were considered for analysis as depicted in Table A1 in the supplementary document. With a 20% temporal step and 10% daily random fluctuation, the total anticipated demand, which is the synthetic load used in the HOMER software tool, is 2,822.2 kWh/day. Figure 2 displays the community's daily and yearly baseline load profiles produced by the HOMER software tool. The Figure portrays varying demand for electricity, ranging from low demand in the early morning hours (12:00 am - 4:00 am) when residents are sleeping to significant spikes in the evening

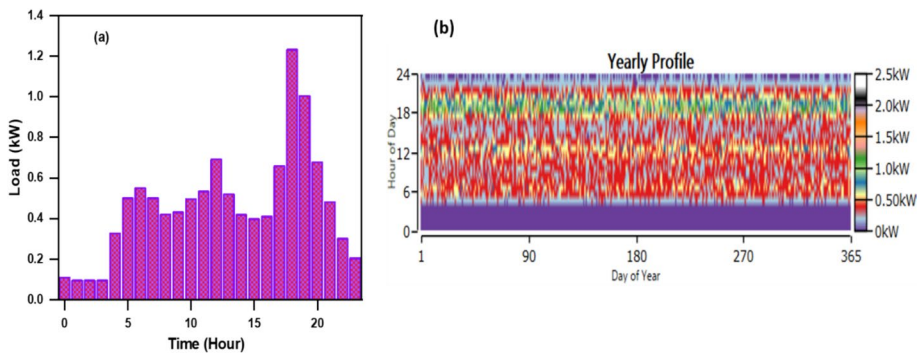


Fig. 2 Giri community electricity demand (a) Daily demand (b) Seasonal Demand

hours (5:00 pm - 9:00 pm) when everyone is back home from various engagements, with most appliances put to use.

2.3 Energy sources

2.3.1 Hydro resources

The Giri community has a run-of-river (River Wuye), which is used predominantly for fishing and irrigation. It navigates the central Campus of the University of Abuja at Giri and discharges into the Usuma River, a tributary of the Gurara River in Niger State, Nigeria. A feasibility study on harnessing the River for mini-hydroelectric power generation by Ozigis et al. (2020) revealed its enormous electricity generation potential, including minimal impacts on the habitats and ecology that may result from changes in the topography and river flow regime. Table 1 shows the hydrological profiles, while the monthly flow rates of the River are presented in Fig. 3(a). As seen in the Figure, the river experiences minimal flow every year in the dry season from November to April and peak flow from July to October in the rainy season.

Table 1 Hydrological profiles of River Wuye (Ozigis et al., 2020)

Parameters	Values
Run-off from rain (mm)	1.30
Dam height (m)	10
River Cross-sectional Area (m ²)	16.90
Gross head (m)	11
Net head (m)	10
Head loss (m)	1
Maximum core height (m)	8
Speed of the river m/s	0.86
Dam length (m)	120
Reservoir (m)	4,000
Flow rate of (L/s)	1,454

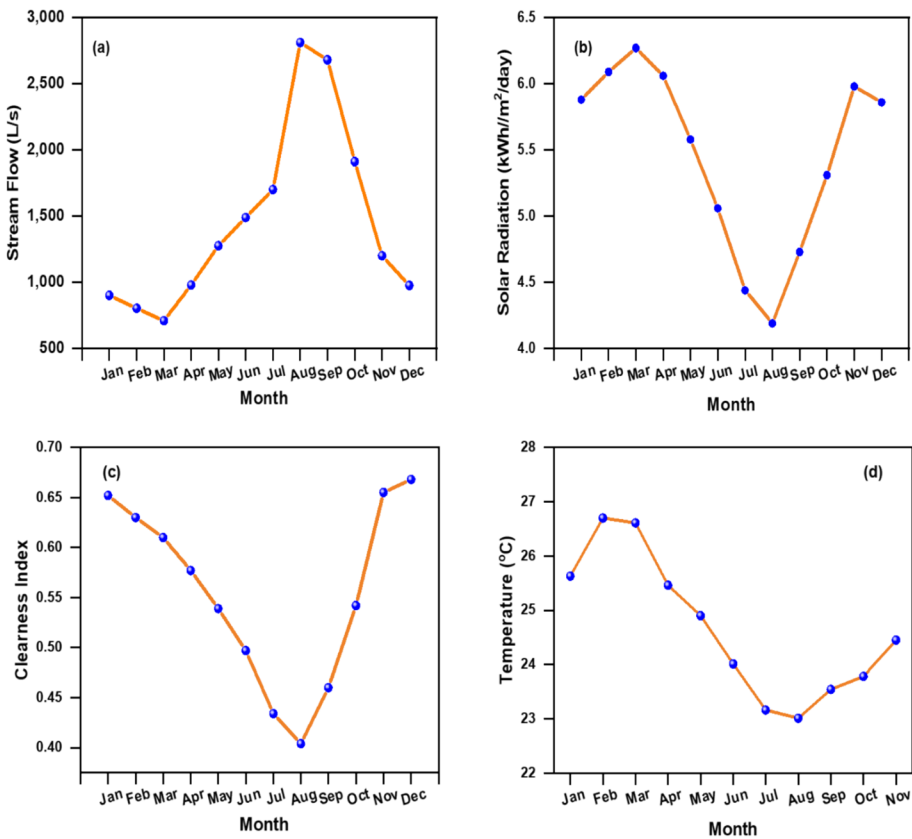


Fig. 3 Giri Community renewable resources (a) Monthly river flow rates, (b) Monthly mean solar radiation, (c) Monthly mean clearness index, (d) Monthly mean temperature

2.3.2 Solar resources

The major solar resources for PV modelling in HOMER software include the clearness index, global horizontal irradiance (GHI), and temperature. The monthly profiles of these variables, shown in Fig. 3(b)–(d), were downloaded from the NASA database. The clearness index indicates the share of solar radiation that hits the Earth's surface after it traverses the atmosphere. Thus, the clearness index has a high value when the weather is clear and sunny, and has a low value when the weather is cloudy. Direct normal irradiance, diffuse horizontal radiation, and ground-reflected radiation are all included in the GHI, which represents the entire amount of solar energy absorbed on a horizontal surface (Dodo et al., 2024). HOMER uses solar GHI to compute the PV output. Meanwhile, the temperature of a PV cell, which is a function of the ambient temperature, has an enormous influence on the output and overall efficiency of a PV array. Thus, a higher operating temperature results in a drop in PV array efficiency. As seen in Fig. 3(b)–(d), besides the run-of-river, the Giri community is endowed with solar resources that can be harnessed for sustainable power generation. The GHI, clearness index, and temperature are at their peaks in March, December, and February, respectively.

2.4 Power systems components modelling

2.4.1 Micro-hydropower plant

The hydraulic turbines generate electricity by converting the kinetic energy of falling water into mechanical power, which a generator subsequently transforms into electrical power. As the first step in the mathematical representation of a micro-hydropower plant, the discharge of the water from a water source is computed according to the catchment area, defined as an area where rain falls towards the anticipated source of water (Syahputra & Soesanti, 2021). This is expressed in Eq. (3).

$$Q_a = K_s Q_g \left(\frac{A_a}{A_g} \right) \quad (3)$$

where Q_a is the discharged water at the site of the micro-hydro power plant (m^3/s), K_s represents the constant of scaling, A_a represents the hydro-power plant's water catchment area (m^2), A_g represents the gauge's catchment area (m^2), Q_g is the water discharge in the gauge (m^3/s). HOMER uses Eq. (4) to compute the hydro turbine power generation (Oladigbolu et al., 2020).

$$P_h = \eta_h \times \rho_w \times h_{net} \times Q_t \times g \times 10^{-3} \quad (4)$$

where g is the gravitational acceleration (m/s^2), η_h is the hydro turbine efficiency (%), Q_t refers to the stream flow rate of the hydro turbine in m^3/s , ρ_w is the density of water ($1000 \text{ kg}/\text{m}^3$), and h_{net} is the effective head in meters, which is computed using Eq. (5) (Oladigbolu et al., 2020).

$$h_{net} = h(1 - f_h) \quad (5)$$

where f_h and h , respectively, denote the pipe head loss and the available head in meters. A circular pipe dominates the application in water conveyance to the turbine. Hence, with an assumption of a circular pipe, the Darcy–Weisbach in Eq. (6) is utilized to calculate the frictional losses (Oladigbolu et al., 2020).

$$h_L = f_D \left(\frac{L}{D} \right) \left(\frac{v^2}{2g} \right) \quad (6)$$

where V is the flow velocity (m/s), f_D is the friction factor of the Darcy, h_L represents the absolute head loss resulting from the friction (in units of length), the pipe length in meters is denoted by L , while the pipe diameter (m) is represented by D .

The techno-economic characteristics of the turbine used in the design are presented in Table B1 in the Supplementary document. The intake pipe loss of this turbine is 15%, while the turbine's available head, design flow rate, minimum flow ratio, maximum flow ratio, and efficiency used in simulation, respectively, are 25 m, 500 L/s, 50%, 150%, and 80%. The River Wuye, which will be channelled to the powerhouse, experiences peak flow from July to October in the rainy season and minimal flow in the dry season from November to April

each year. Therefore, it is essential to reinforce the equipment and civil works in the rainy season because of high seasonal flow changes. In addition to controlling floods, fine aggregates may escape into the turbine systems, increasing wear rates and maintenance costs for the hydropower plant. As a result, desilting the power plant regularly will be required. Furthermore, the hydrological characteristics of the River Wuye showed that the chosen turbine capacity could be maintained year-round without reservoir storage while generating base load steadily (Ondachi, 2024; Ozgis et al., 2020).

2.4.2 Photovoltaic model

Photovoltaic technology (PV) is an innovative technique for capturing solar energy and turning it into electricity. Fundamentally, PV is based on the idea of the photovoltaic effect, which states that when exposed to sunlight, semiconductor materials produce an electric current. This current is in the form of DC, which can be stored or inverted for an AC application. In this study, any excess energy generated by the PV is applied to charge the battery bank, which may then be used to power the necessary load, particularly at night when the sun cannot shine. The PV module's non-linear characteristics of the voltage-current relationship can be expressed mathematically as (Al-Quraan & Al-Mhairat, 2024; Vu et al., 2019):

$$I_{pv} = (I_{ph}N_{p,pv} - I_{rs}N_{p,pv}) \times \left[\exp \left(\frac{q(R_s I_{pv} + V_{pv})}{K T_c N_{s,pv} A_c} \right) - 1 \right] \quad (7)$$

where V_{pv} represents the solar PV module voltage, I_{ph} is the generated current according to irradiance, and I_{rs} stands for cell reversal saturation current. The variables T_c , q , A_c , R_s , K , $N_{s,pv}$, and $N_{p,pv}$ represent the cell's temperature, electron charge, constant of the p–n junction, intrinsic cell resistance, Boltzmann constant, number of cells in series, and number of cells in parallel per string. I_{rs} and I_{ph} , which depend on the irradiance and surrounding temperature, are defined by eqs. (8)–(10) as (Al-Quraan & Al-Mhairat, 2024):

$$I_{rs} = I_{ph} \left(\frac{T_c}{T_r} \right)^3 \times \exp \left(\frac{E_g q \left(\frac{1}{T_r} - \frac{1}{T_c} \right)}{K A_c} \right) \quad (8)$$

$$I_{ph} = (I_{sc} - K_i(T_r - T_c)) \times 0.001\lambda \quad (9)$$

$$T_c = T_a + \lambda \times \left(\frac{NOCT - 20}{800} \right) \quad (10)$$

where K_i is the temperature coefficient of short circuit current. The short circuit current of the cell is considered with reference to irradiance and temperature in kelvin (T_r). The cell's band-gap energy is represented by E_g , and I_{or} is the reverse saturation current based on the reference temperature. The operating cell's nominal temperature ($^{\circ}\text{C}$) is represented as NOCT. Therefore, the aggregate PV power (P_{pv}) for a given quantity of PV arrays ($N_{p,pv}$) can be computed using Eq. (11) (Adetoro et al., 2023; Dodo et al., 2024).

$$P_{pv} = N_{p,pv} \times p_{pv,m} \quad (11)$$

where $P_{pv,m}$ is the peak power that can be achieved by the maximum power point tracking technique. The techno-economic characteristics of the solar PV panels employed in the analysis are presented in Table B2 in the Supplementary document. The capacity of the panel was chosen to fulfil peak demand during the day and complement hydro output during times of low river flow. This sizing was based on the available average solar irradiation of $5.2 \text{ kWh/m}^2/\text{day}$ in the area (Adamu & Dodo, 2025).

2.4.3 Battery model

A battery is a collection of electrochemical cells linked either in parallel (to sustain a certain voltage and improve the capacity) or in series (to raise the output voltage value), depending on the voltage and current characteristics needed for a given application. Battery energy storage systems (BESS) are essential for utilities, businesses, and homes to achieve continuous power flow. They are crucial backup components to energy systems, particularly those that rely on renewable sources or utility grids experiencing recurrent interruptions. The rating of BESS, generally specified in watt-hours (C_{wh}) and ampere-hours (C_{Ah}), is crucial to analysing the reliability of the system. The mathematical expression for C_{wh} of the BESS is depicted in Eq. (12) (Adetoro et al., 2023; Salisu et al., 2019).

$$C_{wh} = \frac{(AD \times E_L)}{(DoD \times \eta_{bat})} \quad (12)$$

where DoD is the discharge depth of the battery, E_L is the mean daily load (kWh/day), AD represents the daily autonomy of the battery, and η_{bat} represents the battery efficiency. The battery state of charge (SoC) is expressed in Eq. (13) (Yadav et al., 2024).

$$SoC(k+1) = SoC(k) + \frac{P_{bch} - P_{bdisch}}{n_b E_{b,u}} \Delta t \quad (13)$$

where P_{bch} and P_{bdisch} denote the powers in the battery in charging and discharging modes, respectively. $E_{b,u}$ is the rating of each battery string, and n_b represents the strings of the battery. The battery can be discharged up to the minimum level (SoC_{min}), analysed based on the depth of the battery's discharge (DoD) as depicted in Eq. (14) (Yadav et al., 2024).

$$SoC_{min} = (1 - DoD) \times C_{bat} \quad (14)$$

C_{bat} expresses the capacity of the battery. The techno-economic characteristics of the battery utilised in the analysis are presented in Table B3 in the Supplementary document. The selected battery capacity was envisaged to balance autonomy and cost, ensuring backup power during dry-season flow reduction and grid unavailability.

2.4.4 Converter

The converter regulates the voltage and frequency to optimize the electricity distribution and seamlessly incorporate BESS, hydro, and solar energy plants into the grid. It functions

as a rectifier and an inverter to synchronize electricity flow across AC and DC buses. The governing expression of the process is shown in Eq. (15) (Adetoro et al., 2023).

$$P_{ac} = P_{dc} \times \eta_{inv} \quad (15)$$

where P_{ac} , P_{dc} , and η_{inv} respectively, represent the AC output power (Watts), DC input power (Watts), and the efficiency of the inverter (%). The techno-economic characteristics of the converter utilised in the analysis are presented in Table B4 in the Supplementary document. The converter is scheduled to operate for 10 years at 96% inversion efficiency and 94% rectification efficiency. The converter capacity was chosen to minimize conversion losses and ensure operational flexibility by taking into account the maximum anticipated simultaneous power exchange between the AC and DC buses.

2.4.5 Grid

The grid tariff in Nigeria, known as "service reflective", does not include net metering, and the unit cost of electricity is uniform regardless of the peak and off-peak periods. The electricity consumers are assigned different Bands from A-D depending on the duration of hours of steady electricity they are scheduled to receive from the grid per day. Consumers in Band "A" are expected to have an interruption of not more than four hours per day, while the grid interruptions for those in Bands "B" and "C", respectively, should not be more than 8 and 12 h per day. At the bottom of the ladder are bands "D" and "E," which are scheduled for minimum grid interruptions of 8 h per day. Meanwhile, this structure is currently impractical as those in Band "A" grapple with incessant interruptions that last for hours or even days. Therefore, the Schedule Rates in the Advanced Grid component of HOMER Pro were used to model an unreliable grid. Grid interruptions occurring twice and lasting for an average of 6 h per day, with 90% variability in repair time, are typical of customers in Band A in Nigeria. The grid interruptions Dmap produced by the HOMER program using these data are shown in Fig. 4. In the graphic, the grid's regular operation is shown in green, while black patches indicate when there is a grid interruption.

The characteristics of the grid emission employed in the simulation are based on the electricity mix of Nigeria, which comprises 378 g/kWh of CO₂, 0.41 g/kWh of NO_x, 1.9 g/kWh of SO₂, 0.03 g/kWh of CO, and 0.14 g/kWh of particulate matter (PM) (Dodo et al., 2024; Jumare et al., 2020). Additionally, Band A, in which the research location falls, has a grid tariff of \$0.18 per kWh. Equation (16) expresses the yearly cost of purchasing electricity from the grid.

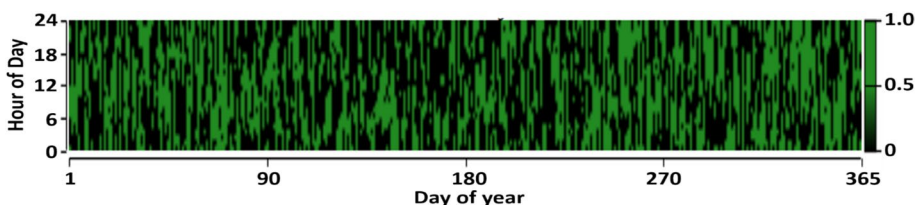


Fig. 4 Giri grid outage characteristics

$$C_g = \sum_{t=1}^T P_{g,t} \times \beta \quad (16)$$

where C_g is the unit cost of grid electricity (\$/kWh), $P_{g,t}$ represents the electricity imported from the grid at an hour t (kW), while T represents the study interval's total duration (1 to 8760 h for a year).

2.5 System economics

The attainment of the minimal values of LCOE and the NPC, which are regarded as the most important economic indicators, characterizes an optimal hybrid energy system (Al-Quraan & Al-Mhairat, 2024; Yadav et al., 2024). Hence, the simulation and optimization were performed in HOMER to minimize the NPC and LCOE. The system is scheduled to have a zero capacity shortage as the constraint.

2.5.1 Net present cost

To assess the NPC, the annual inflation rate (I_f) and nominal interest rate (I_n) are considered according to the country in which the power plants are sited. In HOMER, the NPC is evaluated using Eq. (17) (Yadav et al., 2024).

$$NPC = \sum_{P_L=1}^{P_L=25} R_d(C_{rep} + C_{cap} + C_{OM} - C_{sv}) \quad (17)$$

where C_{cap} is the components' capital cost, C_{rep} is the cost of replacement of the components, C_{OM} is the cost of O&M of the components, P_L specifies the project lifetime, and C_{sv} represents the salvage value. In this study, the project is planned to have a lifetime of 25 years.

The discount rate (R_d) and the annual real interest rate (I_r) are computed using eqs. (18) and (19), respectively (Salisu et al., 2019; Yadav et al., 2024).

$$R_d = \frac{1}{(I_r + 1)^{P_L}} \quad (18)$$

$$I_r = \frac{I_n - I_f}{(I_f + 1)} \quad (19)$$

I_n and I_f respectively, represent the annual nominal interest rate and inflation rate. Nigeria's inflation rate is 34.60%, while the annual interest rate is 27.5% according to the Central Bank of Nigeria.

The total annualized cost of the HRES components (TAC) is evaluated from the NPC using the capital recovery factor as shown in Eq. (20) (Yadav et al., 2024).

$$TAC = NPC \times CRF(I_r, P_L) \quad (20)$$

The CRF is expressed in Eq. (21)

$$CRF(I_r, P_L) = \frac{I_r(1 + I_r)^{P_L}}{(1 + I_r)^{P_L} - 1} \quad (21)$$

2.5.2 Levelized cost of energy

The levelized cost of energy (LCOE) is one of the key economic indicators of hybrid power systems. It indicates the price per unit of useful electricity produced by the various generating plants. It is expressed as the ratio of the total annualized cost to the total yearly electrical energy supplied to the load, as depicted in Eq. (22) (Salisu et al., 2019).

$$LCOE = \frac{TAC}{E_{tot}} \quad (22)$$

2.5.3 Salvage cost

The cost of a power system component that remains functional at the end of a project's lifetime is expressed as salvage cost. HOMER uses Eq. (23) to evaluate the worth of every component after the project's lifespan (Riayatsyah et al., 2022).

$$S = C_{rep} \frac{R_{rem}}{R_{comp}} \quad (23)$$

C_{rep} , R_{rem} , and R_{com} , respectively, stand for the component replacement cost, the remaining lifetime of the component, and the component's entire lifetime.

2.6 Environmental emission model

Renewable fraction (RF) is a function signifying the degree of RE penetration and can also serve as an indicator of GHG emissions by hybrid energy systems. A system that has a high RF is generally preferred since it will have minimal GHG emissions. The RF is expressed in Eq. (24) (Salisu et al., 2019).

$$RF(\%) = \left(1 - \frac{\sum E_{FG}(t)}{E_{hyd}(t) + E_{PV}(t)} \right) \times 100 \quad (24)$$

where $E_{hyd}(t)$, and $E_{PV}(t)$ respectively, are the fossil-based generators, wind turbines, and solar plants' output energy. Since Nigeria's power grid relies mainly on natural gas and coal for electricity production, the proposed power systems, being grid-connected, will impact the environment negatively. Thus, when fossil fuels are used to produce electricity, a variety of hazardous pollutants are released, including NO_x , PM, carbon monoxide (CO), CO_2 , sulfur dioxide (SO_2), and unburned hydrocarbons (UH). For the sake of environmental sustainability, preference is thus given to the energy systems with the lowest footprints of

these pollutants, with carbon dioxide accorded more attention. The yearly emissions of CO₂ are computed using Eq. (25).

$$A_{CO} = \sum_{t=1}^{8760} CO_2(t) \quad (25)$$

The CO₂ and other emission variables in the current investigation are entirely operational-related. Embedded emissions from the production of PV panels, batteries, and run-of-river system civil works were excluded from this study.

3 Results and discussion

This study conducted techno-economic feasibility analyses of hybrid energy systems leveraging run-of-river and solar resources for a rural community impacted by frequent utility grid outages. Using the HOMER pro software tool, an optimal electrification scheme for the community was explored based on four distinct scenarios, namely, Grid/PV/Hydro, Grid/Hydro, PV/Hydro, and Grid/PV systems. A total of 2,036 solutions were simulated, out of which 1,521 cases were found infeasible due to capacity shortage constraints, and only 515 cases were feasible. Similarly, 42 and 40 cases, respectively, were omitted due to a lack of converter and power generation sources, while 108 cases were omitted for having unnecessary converters. The hybrid energy system, having the least NPC and LCOE, and which can reliably meet the electricity demands of the community, is ranked as the optimal system.

3.1 Economic analysis

The key economic metrics employed for the appraisal of the four energy scenarios included NPC, operating cost, LCOE, and salvage cost, as shown in Table 2. The Grid/PV/Hydro presents the least total NPC of \$8,160,309.00, an operating cost of \$99,923.27/year, an LCOE of \$0.13/kWh, and a salvage cost of \$3,857,031.29 while the Grid/PV has the highest NPC of 40,102,250.00, the operating cost of \$525,520.90/year, an LCOE of \$0.71/kWh, and a salvage cost of \$40,102,246.21. The Grid/PV system's total NPC, operating cost, and LCOE are higher in multiples of 4.91, 5.53, and 5.26 compared to the Grid/PV/Hydro system. Similarly, the PV/Hydro and Grid/Hydro energy systems, respectively, are relatively cost-effective compared to the Grid/PV system. Thus, the NPC, operating cost, LCOE, and salvage cost of the Grid/PV system are at least 3.5 times those of the PV/Hydro and Grid/Hydro systems, respectively. Figure 5 further shows the contribution of the various cost

Table 2 Systems economics

Energy systems	Total NPC (\$)	LCOE (\$/kWh)	Operating cost (\$/year)	Salvage cost (\$)
Grid/PV/Hydro	8,160,309.00	0.13	99,923.27	3,857,031.29
PV/Hydro	9,858,722.00	0.18	117,581.10	4,644,277.14
Grid/Hydro	12,158,800.00	0.20	150,062.10	6,270,448.23
Grid/PV	40,102,250.00	0.71	525,520.90	40,102,246.21

components to the total NPC and the annualized system costs. In both scenarios of the NPC and the annualized system cost, the Grid/PV/Hydro demonstrates a cheaper capital cost, replacement cost, and O&M cost compared to other energy systems. Conversely, the Grid/PV system has the highest salvage cost resulting from the battery (\$17,411,531.36) and converter (\$2,422,433.68).

The total NPC associated with the power components (converter, battery, MHT, solar panels, and grid) is not uniform across the examined systems, except for the MHT, as shown in Fig. 6. The MHT has an NPC of \$1,211,954.84 in Grid/PV/Hydro, PV/Hydro, and Grid/Hydro systems, respectively. The variation in the costs of PV panels, converters, grid, and BESS can be attributed to the fluctuation of energy production from solar PV arrays and electricity importation from the grid. Of the four energy schemes, BESS has the highest NPC, similar to Dodo et al. (2022). It is pertinent to note that the chemistry of the battery employed is lead-acid. Lead-acid batteries have relatively low initial capital cost, but due to the demerit of short lifespan, their replacement costs are always high. Furthermore, the NPC of the converters in Grid/PV/Hydro, Grid/Hydro, and PV/Hydro systems is constant (\$784,155.56) but increased by multiples of 8.33 in the Grid/PV system.

3.2 Technical analyses

3.2.1 Capacities of the components

The capacities of the components in the hybrid energy systems are presented in Table 3. The strings of battery required in parallel are 2,818, 3,869, 4,725, and 13,758, respectively, in Grid/PV/Hydro, PV/Hydro, Grid/Hydro, and Grid/PV systems. Since the grid is unreliable and solar resources are prone to fluctuation and randomness, it becomes imperative to employ large strings of batteries in the energy systems comprising solar and grid only (Grid/PV system). The capacities of the PV panels (100 kW) are the same in Grid/PV/Hydro, PV/Hydro, and Grid/PV. Similarly, the capacities of the MHT (98.1 kW) are uniform in Grid/PV/Hydro, PV/Hydro, and Grid/Hydro systems, respectively. The sizes of the converter do not vary except in Grid/PV systems, in which a large quantum of energy is expected to be stored in the battery for subsequent inversion to enable AC load utilization.

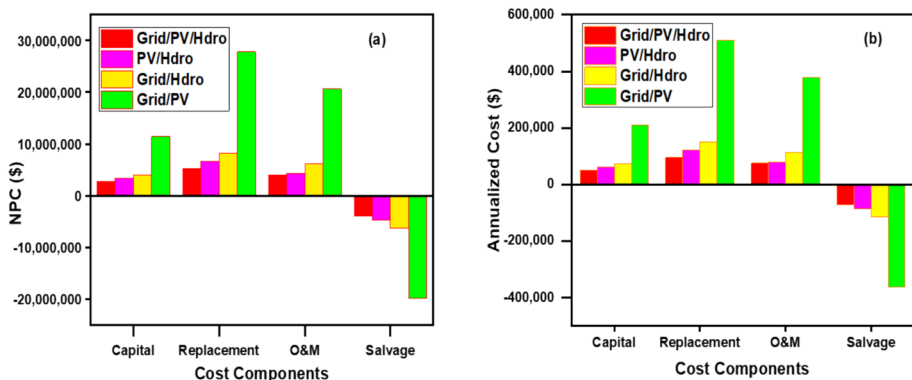


Fig. 5 System economics (a) Total NPC (b) Total annualized cost

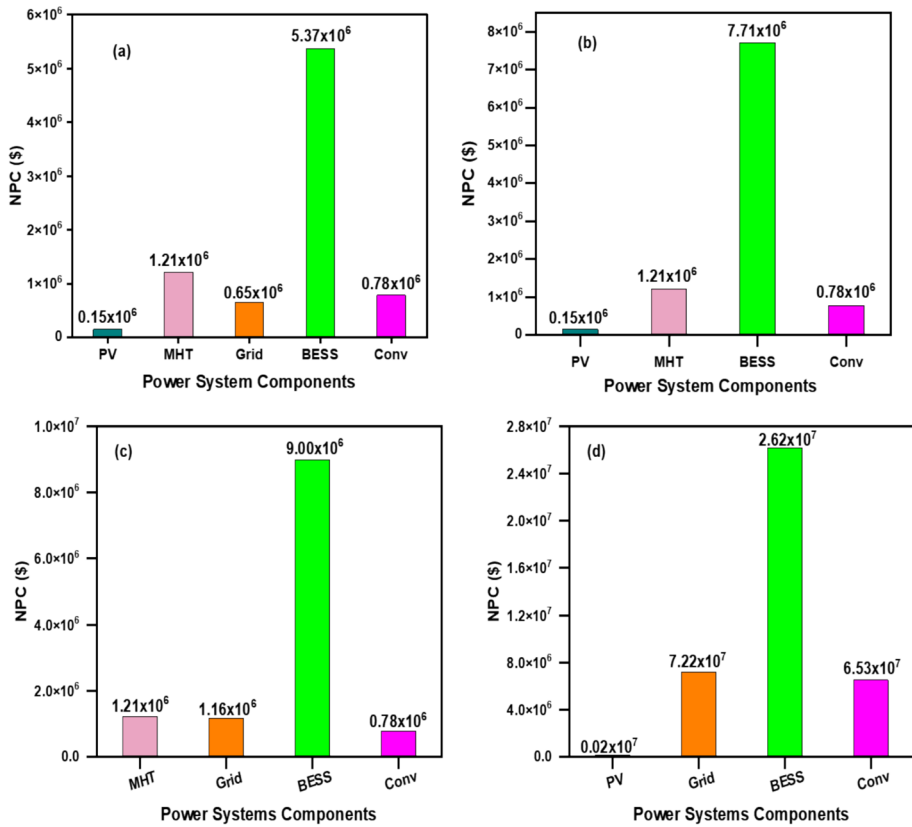


Fig. 6 System components NPC (a) Grid/PV/Hydro system, (b) PV/Hydro system, (c) Grid/Hydro system, (d) Grid/Hydro system

Table 3 Capacities of the HRES components

Energy systems	BESS (Strings)	PV panels (kW)	MHT (kW)	Converter (kW)	Grid (kW)	Dis- patch strategy
Grid/PV/Hydro	2,818	100	98.1	300	999,999	LF
PV/Hydro	3,869	100	98.1	300	-	CC
Grid/Hydro	4,725	-	98.1	300	999,999	CC
Grid/PV	13,758	100	-	2500	999,999	CC

The optimal dispatch strategy for the Grid/PV/Hydro system is load following (LF), while the PV/Hydro, Grid/Hydro, and Grid/PV systems, respectively, utilize the cycle charging (CC) strategy for efficient energy management. In an LF strategy, the solar PV and micro-hydro generators only provide electricity sufficient to satisfy the existing load demand, without necessarily charging the battery to its maximum capacity, while in a CC strategy, the solar PV and micro-hydro generators operate at full capacity whenever necessary, charging the battery with any excess electricity. In other words, LF prioritizes the gen-

eration of power required to satisfy the immediate demand, while the CC strategy prioritizes charging the battery fully whenever possible.

3.2.2 Electricity generation

A total of 1,348,846 kWh is produced annually by the Grid/PV/Hydro system, while the PV/Hydro system, Grid/Hydro system, and Grid/PV system, respectively, produce 1,257,681 kWh/year, 1,254,969 kWh/year, and 1,185,283 kWh/year. The breakdown of the contribution to the pool of electricity by the various energy sources is shown in Fig. 7. As shown in the Figure, hydro-power plants account for a substantial share of electricity generation, while the grid is rarely relied upon due to its unreliable nature, except in Grid//PV system, where it dominates electricity production. Thus, in the Grid/PV/Hydro system, about 80.9%, 12.4%, and 6.7% of the total electricity, respectively, is supplied by hydro-power plants, solar PV arrays, and the grid. Similarly, in the Grid/Hydro system, the hydro-power plant is responsible for 86.9%, while the grid supplies the remaining 13.1% of the total electricity demand of 1,254,969 kWh/year. The Grid/PV system has no excess electricity generation compared to Grid/PV/Hydro, PV/Hydro, and Grid/Hydro systems, having 152,017 kWh/year, 162,953 kWh/year, and 96,114 kWh/year, respectively. The excess generation,

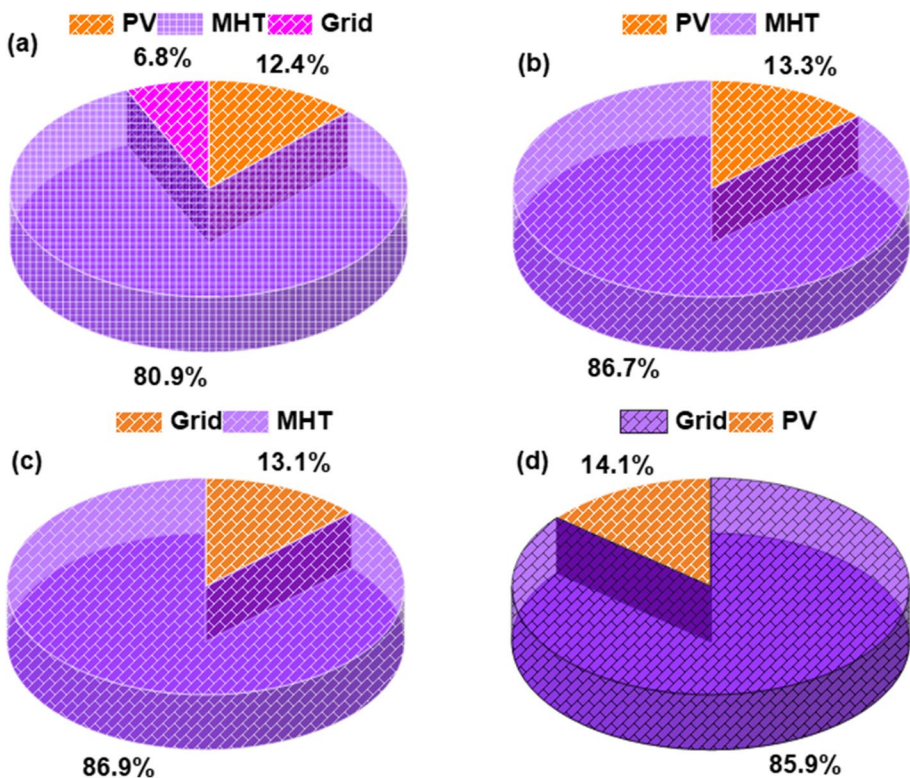


Fig. 7 Share of electricity supply (a) Grid/PV/Hydro system (b) PV/Hydro system (c) Grid/Hydro system (d) Grid/PV system

which can serve deferrable loads, is attributed to the dominance of hydro-power plants in the share of energy production.

The monthly average electricity production for the studied cases is presented in Fig. 8. The run-of-river potential for electricity generation is dominant throughout the year in hybrid Grid/PV/hydro, PV/hydro, and PV/Hydro systems, respectively. The maximum electricity generation by hydropower plants is 93,057.66 kW. This occurs in July and August when the seasonal rainfall is at its peak. Meanwhile, the lowest generation of electricity by the hydropower plants occurs in February (84,052.08 kW) and March (87,984.25 kW), when the dry season has fully set in, corresponding to the period of high solar penetration. In the Grid/PV system, the community's electricity demand is met substantially by the utility grid. Thus, the system supports electricity supply from the grid over PV arrays. This can be attributed to the volatile and fluctuating nature of solar resources (Dodo et al., 2024).

3.2.3 Battery performance

Battery autonomy, energy-in, energy-out, throughput, storage wear cost, storage depletion, expected life span, and losses are amongst the prominent and useful indices for assessing the BESS performances. The values associated with these metrics are presented in Table 4. The battery autonomy is the longest period of continuous energy release by the battery. It is the amount of time that the load can be satisfied solely by the battery without the need for hydro, solar, or grid supply. The battery bank can reliably supply the Grid/PV system continuously for 96 h without any input from the PV panels or the grid. Similarly, the Grid/PV/Hydro,

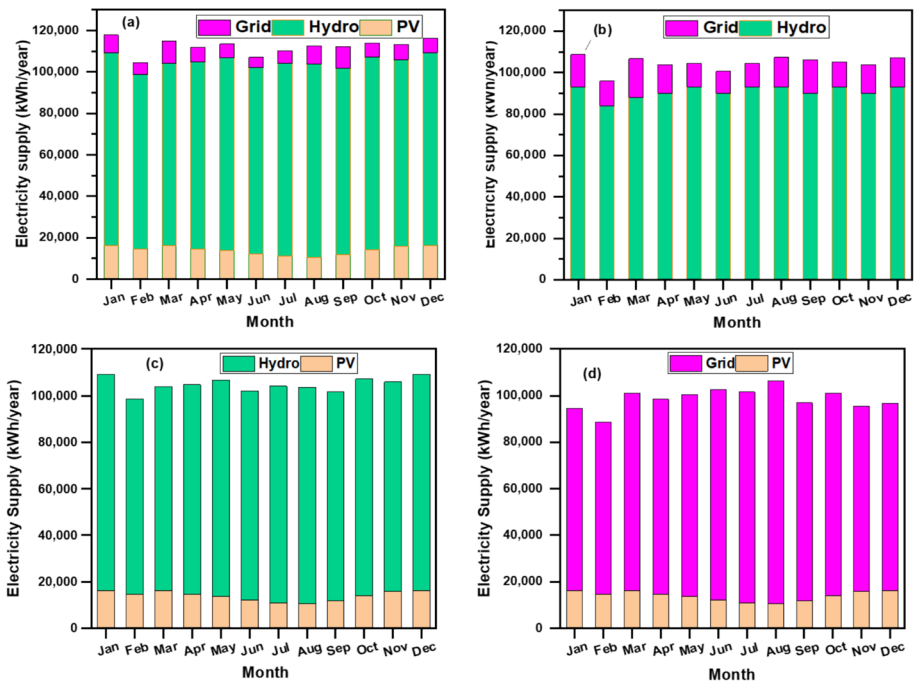


Fig. 8 Monthly average electricity generation: (a) Grid/PV/Hydro system, (b) Grid/Hydro system, (c) PV/Hydro system, (d) Grid/PV system

Table 4 Battery performance analysis

Parameters	Grid/PV/Hydro	PV/Hydro	Grid/Hydro	Grid/PV
Autonomy (hr)	19.70	27.00	33.0	96.00
Storage Wear Cost (\$/kWh)	0.93	0.70	0.952	1.02
Expected Life (yr)	16.70	16.10	16.70	16.70
Energy In (kWh/yr)	126,742	239,476	207,565	572,260
Energy Out (kWh/yr)	106,873	202,480	175,657	478,560
Storage Depletion (kWh/yr)	147	354	455	2,476
Storage Losses (kWh/yr)	20,016	37,350	32,362	96,176
Annual Throughput (kWh/yr)	116,589	220,541	191,349	520,069

PV/Hydro, and Grid/Hydro systems, respectively, have an autonomy of 19.7 h, 27.0 h, and 33.0 h. The energy-in indicates the aggregate energy supplied annually to charge the battery bank, while the energy-out measures the quantum of energy discharged by the battery bank per year. The maximum energy-in of 572,260 kWh/year is achieved for the Grid/PV system, while the Grid/PV/Hydro has the minimum energy-in of 126,742 kWh/year. The yearly charging energy for PV/Hydro and Grid/Hydro, respectively, is 41.8% and 36.3% of the yearly charging energy for Grid/PV systems. The energy-out of the battery banks follows the same pattern as the energy-in.

The entire quantity of energy that flows through BESS during its lifetime in both the charging and discharging phases is known as the throughput. The energy constraint and best scheduling of BESS during the planning period can be established based on the battery's energy throughput and anticipated lifespan. Therefore, the annual throughput of the battery banks for Grid/PV/Hydro, PV/Hydro, Grid/Hydro, and Grid/PV systems, respectively, is 116,589 kWh/year, 220,541 kWh/year, 191,349 kWh/year, and 520,069 kWh/year. The storage depletion which expresses the difference in the battery's SoC between the beginning and end of the year is highest in Grid/PV system (2,476 kWh/yr) followed by the Grid/Hydro system (455 kWh/yr), PV/Hydro system (354 kWh/yr), and Grid/PV/Hydro system (147 kWh/yr). It is impractical to have a loss-free energy storage device. Thus, the battery bank loses some quantum of energy during discharging due to storage inefficiencies. The storage depletion and the energy loss in kWh/year both indicate a similar pattern with the annual throughput. Wear cost is another important factor for the performance evaluation of storage devices. It describes the cost of a battery's steady capacity loss over time resulting from continuous usage. As the battery ages, it can store less charge and will later need to be replaced, giving rise to a new battery purchase. The storage wear costs associated with the Grid/PV/Hydro, PV/Hydro, Grid/Hydro and Grid/PV systems, respectively, are \$0.93/kWh, \$0.70/kWh, \$0.95/kWh, and \$1.02/kWh.

One of the main causes of a battery's decreased lifespan is the aging and fading of the active components brought on by the cycles of charging and discharging, as well as the operating temperature. Therefore, the battery lifespan can be a crucial performance indicator when it comes to assessing its economic efficiency. The length of time that a battery storage system is anticipated to function following its defined features and meet the required performance standards is referred to as its "expected life." The expected life of the battery is the same (16.7 years), except for the PV/Hydro system, whose expected battery lifespan is 16.1 years. This implies that all the batteries must be replaced before the expiration of the HRES project's lifetime of 25 years.

The SoC of a battery is another important metric for monitoring its health and guaranteeing its safe operation. The percentage of the battery's capacity that can be discharged to the loads is known as the SoC. The SoC is regarded as 100% for a fully charged battery and 0% for an entirely depleted battery. The Dmaps for the SoC of the four energy systems are shown in Fig. 9. The SoC is irregular throughout the year. For instance, on May 11th, the Grid/Hydro system attained a maximum SoC of 87% from 7:00 am – 8:00 am and further declined to a minimum of 67% at 7:00 pm. Meanwhile, on May 16th, a 100% SoC is observed from 12:00 am to 9:00 am before descending to a minimum of 85% at 6:00 pm. In the Grid/PV/Hydro system, the SoC rose from 80% at 12:00 am to 95% at 8:00 am of May 11th and then continued to decay up to a minimum of 72% at 7:00 pm. Similarly, on May 16th, a 100% SoC is attained from 12:00 am to 9:00 am before a sharp decrease to a minimum of 95% at 9:00 pm. The Grid/PV system's BESS is characterized by the lowest mean SoC of 82%, while the Grid/PV/Hydro system's BESS has the highest mean SoC of 98%. The mean peak battery SoC for the Grid/PV system is 84%, implying that throughout the lifespan of the project, its battery bank cannot attain an SoC above 84% on average.

3.3 Emission analyses

The emission characteristics of the investigated hybrid energy systems are presented in Table 5. The extent of penetration of solar and hydro resources in relation to the grid and fossil fuels is designated as the renewable fraction (RF). Due to the inclusion of grid emission characteristics during the modeling (Dodo et al., 2024), the grid-connected systems are unable to attain an RF beyond 92.1% (Grid/PV/Hydro). The worst case is observed in Grid/PV, which has an RF of 1.10%, implying its substantial negative environmental impacts. This is further corroborated by other emission variables such as UH, CO₂, CO, PM, NO, and SO₂. For instance, the annual CO₂ emission of Grid/PV/Hydro and Grid/Hydro systems, respectively, is 8.95% and 16.14% of the annual CO₂ emission associated with the Grid/PV system (384,882 kg/yr). Since the PV/Hydro system is off-grid and utilizes run-of-river and solar resources only, it is considered the cleanest energy system compared to the Grid/PV, Grid/Hydro, and Grid/PV//Hydro systems. More so, the high rates of emissions associated with grid-connected systems are due to the inclusion of the Nigerian grid emission characteristics during simulations in the HOMER software tool (Dodo et al., 2024; Jumare et al., 2020).

The annual CO₂ of 218,497 kg and 148,000 kg, respectively, are emitted by the Grid/PV/WT/DG (Al-Rawashdeh et al., 2023) and PV/WT/DG (Roy et al., 2024). Thus, these are huge values compared to the emissions of the Grid/PV/Hydro, PV/Hydro, and Grid/Hydro energy systems of this study. In Bayazit (2021), a cursory look at the GHG emissions of dif-

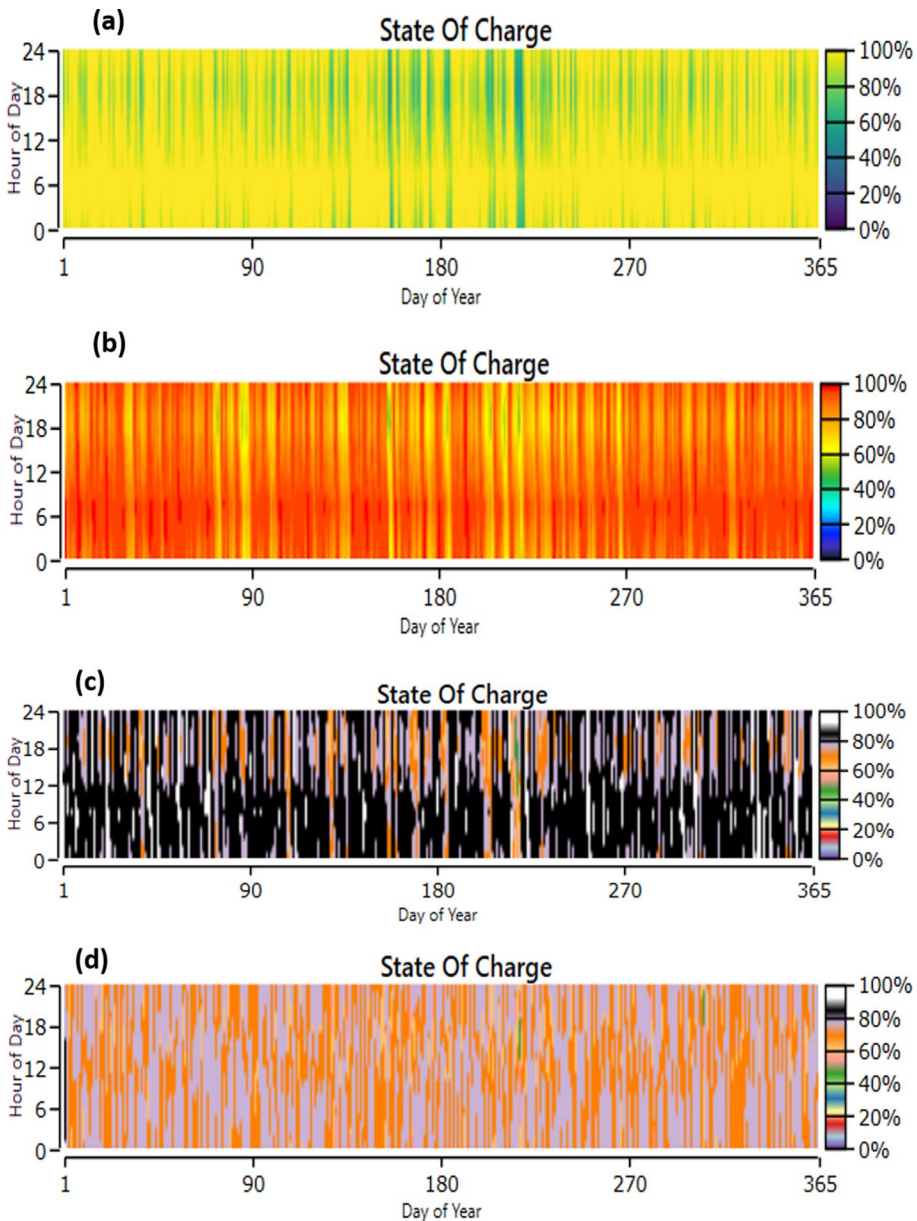


Fig. 9 SoC of the batteries: (a) Grid/PV/Hydro system, (b) PV/Hydro system, (c) Grid/Hydro system, (d) Grid/PV system

ferent energy plants was taken. Hydropower plants are reported to have the minimum GHG emissions per kilowatt-hour. Their cycle carbon equivalent density is 18.50 gCO₂-eq/kWh, whereas a coal power plant's is 820 gCO₂-eq/kWh. The study further acknowledged that hydropower plants can lower GHG emissions by 96.2% per kilowatt-hour of natural gas, 92% per kilowatt-hour of biomass, 61.5% per kilowatt-hour of solar PV, and 51.3% per kilo-

Table 5 Emission analysis

Emission variable	Grid/PV/Hydro	PV/Hydro	Grid/Hydro	Grid/PV
CO ₂ (kg/yr)	34,460	0	62,130	384,882
CO (kg/yr)	2.73	0	4.93	30.50
UH (kg/yr)	0	0	0	0
PM (kg/yr)	12.80	0	23.00	143
SO ₂ (kg/yr)	173	0	312	1,935
NO (kg/yr)	37.4	0	67.4	417
RF (%)	92.10	100	85.10	1.10

watt-hour of geothermal, respectively. This demonstrates the vital role hydro resources can play in curtailing the environmental degradation ascribed to fossil-based energy sources. Therefore, using hydro resources, particularly the run-of-river that traverse various parts of the world, is a guaranteed way of curbing the adverse environmental impacts and an important step towards a cleaner energy future.

3.4 Sensitivity analyses

The sensitivity analysis was performed to provide an overview of the influence of certain variables on the operation of the optimal energy system (Grid/PV/Hydro system). The electricity demand of the project's location will rise because of the envisaged massive infrastructural development. Thus, the community is a suburb of the federal capital city and an off-campus residence for the students of the University of Abuja, Nigeria. More so, the river flow rate is prone to seasonal fluctuation, and the system is not immune to economic uncertainties such as inflation rates. Therefore, it is imperative to visualize the optimal system's responses to the daily load demand, river flow rates, inflation rates, and discount rates. The scaled electricity demand was increased to 4,233.30 kWh/day in steps of 5%, while the inflation rates and river flow rates variations, respectively, are in steps of $\pm 20\%$ and $\pm 30\%$ of their scaled values. Furthermore, a $\pm 2\%$ variation around the base value of 27.50% of the discount rate was applied. Each of these variables affects the optimal energy system in diverse ways, as shown in the component ratings, systems' dispatch strategies, and RF in Tables 6, 7 and 8 and system economics in Fig. 10.

3.4.1 Load demand

In contrast to the battery banks and converters, the capacities of the PV panels and MHT do not change with an increase in the daily electricity demands, as shown in Table 6. The optimal dispatch strategy, with an increase in the daily load demands in steps of 5% up to 50%, remained cycle charging. Meanwhile, the more the electricity demand, the lower the RE penetration, and the more the systems emit harmful gases, as indicated by the decreasing RF.

Table 6 Load demand as a sensitivity variable

Load demand (kWh/day)	PV panels (kW)	Battery (kW)	MHT (kW)	Converter (kW)	Dispatch strategy	RF (%)
2,963.31	100	3,511	98.1	300	CC	90.31
3,104.42	100	3,797	98.1	700	CC	88.24
3,245.53	100	4,818	98.1	700	CC	87.43
3,386.64	100	5,336	98.1	1,200	CC	85.28
3,527.75	100	6,404	98.1	1,200	CC	84.35
3,668.86	100	8,383	98.1	700	CC	83.56
3,809.97	100	9,565	98.1	700	CC	81.88
3,951.08	100	10,747	98.1	700	CC	79.79
4,092.19	100	9,565	98.1	2,500	CC	75.95
4,233.30	100	11,177	98.1	2,500	CC	73.74

Table 7 River flow rate as a sensitivity variable

River flow rate (L/s)	PV panels (kW)	Battery (kW)	MHT (kW)	Converter (kW)	Dispatch strategy	RF (%)
1,017.8	100	3,078	98.1	300	CC	89.45
1,090.5	100	2,914	98.1	300	LF	91.06
1,163.2	100	2,860	98.1	300	LF	91.45
1,235.9	100	2,842	98.1	300	LF	91.71
1,308.6	100	2,842	98.1	300	LF	91.89
1,381.3	100	2,818	98.1	300	LF	92.04
1,454.0	100	2,818	98.1	300	LF	92.14
1,526.7	100	2,818	98.1	300	LF	92.23
1,599.4	100	2,818	98.1	300	LF	92.24
1,672.1	100	2,818	98.1	300	LF	92.24
1,744.8	100	2,818	98.1	300	LF	92.24
1,817.5	100	2,818	98.1	300	LF	92.24
1,890.2	100	2,818	98.1	300	LF	92.24

Table 8 Inflation rate as a sensitivity variable

Inflation rate (%)	PV panels (kW)	Battery (kW)	MHT (kW)	Converter (kW)	Dispatch strategy	RF (%)
27.68	100	2,818	98.1	300	LF	92.14
29.41	100	2,818	98.1	300	LF	92.14
31.14	100	2,818	98.1	300	LF	92.14
32.87	100	2,818	98.1	300	LF	92.14
34.60	100	2,818	98.1	300	LF	92.14
36.33	100	2,818	98.1	300	LF	92.14
38.06	100	2,818	98.1	300	LF	92.14
39.79	100	2,818	98.1	300	LF	92.14
41.52	100	2,818	98.1	300	LF	92.14

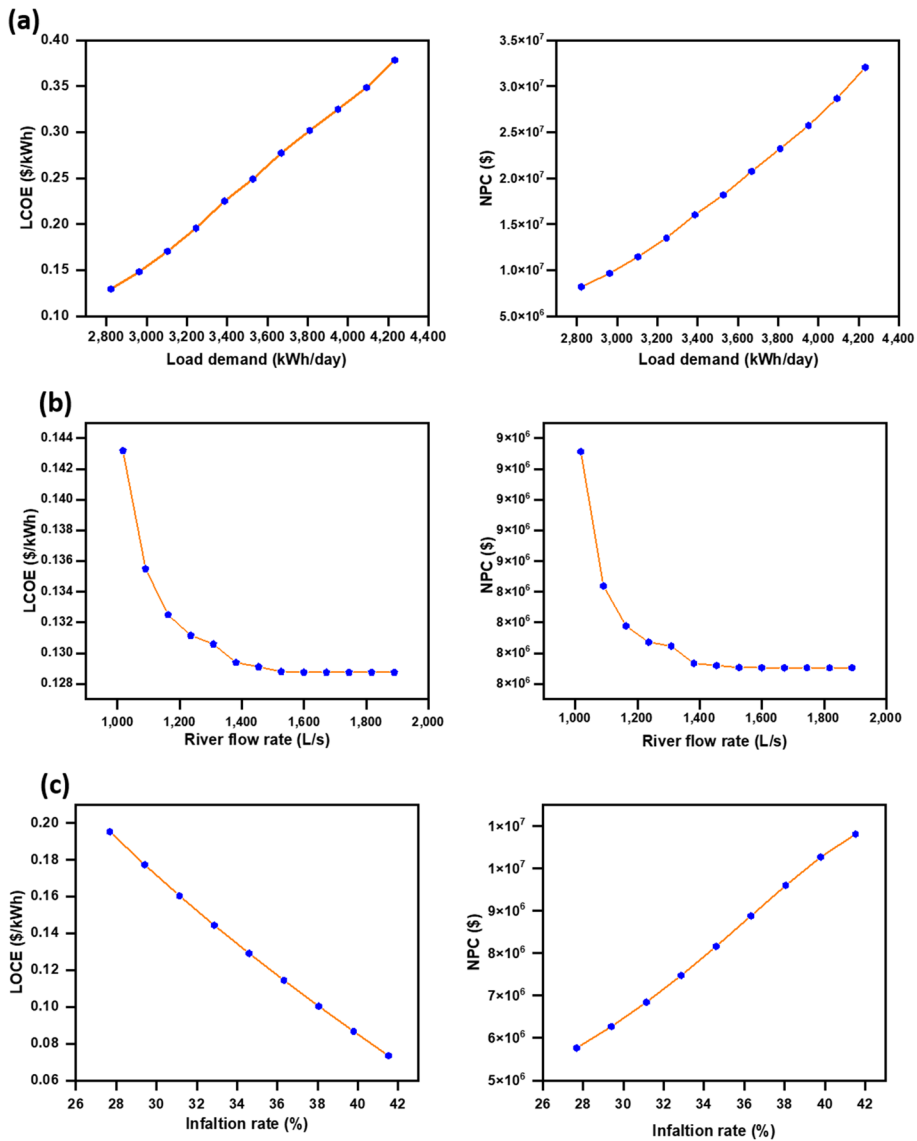


Fig. 10 LCOE and NPC outlook based on the sensitivity variables (a) Load demand, (b) Seasonal flow rate, (c) inflation rate

The LCOE and NPC are vital economic indices to demonstrate the cost-effectiveness of a hybrid energy system. Both the LCOE and NPC increase proportionally with the scaled electricity demand as shown in Fig. 10(a). At a 5% increase in the scaled annual mean demand, the LCOE and NPC, respectively, are \$0.15/kWh and \$9,654,527.00. Meanwhile, when the demand increases by 50%, the LCOE and NPC, respectively, increase substantially to \$0.3786/kWh and \$32,074,934. Thus, the higher the electricity demand, the higher

the NPC and the unit cost of electricity. These incremental system costs result from the increase in the capacities of the components, especially the converters and batteries.

3.4.2 River flow rates

Table 7 depicts that the ratings of the solar PV panels, MHT, and converters are not influenced by the river flow rates, in contrast to the battery banks whose capacities varied by -20% , -25% , and -30% of the scaled river flow rates. The increase in river flow rates causes an increase in the RF, implying decreasing environmental impacts. The load following is the optimal dispatch strategy, except when the river flow rate is decreased by 30% . Conversely, the river flow rates have an inverse relation with the system NPC and LCOE. Meanwhile, an increase in the river flow rates beyond the scaled value of $1,454$ L/s does not translate into lower LCOE and NPC. At -30% of the scaled river flow rates, the NPC and LCOE, respectively, are \$8,857,159.00 and \$0.143/kWh, while at $+30\%$ of the scaled river flow rates, the LCOE and NPC, respectively, are \$0.1288/kWh and \$8,152,774.00.

3.4.3 Inflation rates

Table 8 shows that the inflation rates do not influence the operating strategy and RF of the optimal Grid/PV/Hydro energy system. Thus, despite the variation in the inflation rates, the RF, which can also serve as an indicator of the environmental impacts of the system, remained at 92.14% , while the operating strategy is the LF. Similarly, the capacities of the solar PV panels, batteries, MHT, and converters are unaffected by the inflation rates. In contrast to the electricity demand and the river flow rates, the NPC and LCOE associated with the inflation rates do not have a uniform pattern as depicted in Fig. 10(c). The NPC has a positive linear relationship with the inflation rates, while the LCOE has a negative linear relationship with the inflation rates. For instance, at a 5% increase in scaled inflation rates, the NPC and LCOE, respectively, are \$0.18/kWh and \$6,271,536.00, while at a 40% increase in the scaled inflation rates, the NPC and LCOE, respectively, are 10,805,800.00 and \$0.07/kWh. Nigeria's inflation rates show an annual increasing trend, implying that the community will benefit from a lower LCOE at the disadvantaged NPC.

3.4.4 Discount rates

The findings show that the ideal system configuration is unaffected by changes in the discount rate. As a result, the grid, micro-hydro turbine (98.1 kW), photovoltaic (PV) array (100 kW), converter (300 kW), and battery storage system (2,818 strings) all have the same capacities throughout the evaluated scenarios. Similarly, the load-following mode is maintained as the optimal dispatch technique. The economic indicators of the system, on the other hand, show a discernible sensitivity to changes in the discount rates. The LCOE, total NPC, and yearly operating cost are \$0.12 per kWh, \$8,399,988.00, and \$97,416.05 per year, respectively, using a 27.0% discount rate. These figures change to \$0.13 per kWh, \$7,926,547.00, and \$102,292.30 per year when the rate rises to 27.5% . At 34,460 kg, 2.73 kg, 12.8 kg, 173 kg, and 37.4 kg annually for CO_2 , CO, PM, SO_2 , and NO emissions, respectively, the emissions profile of the optimal system is not impacted under the sensitivity sweep. Additionally, the system logs zero UH, demonstrating its dependability.

3.5 Seasonal dispatch

The hybrid Grid/PV/Hydro system's 48-h dispatch performance and battery SoC profile are shown in Fig. 11 during the typical dry-season period (March), which is marked by lower river discharge. As the system's baseload generator, the MHT produced a steady and consistent output of about 118 kW during the observed period. This regular contribution demonstrates how dependable the run-of-river hydro plant is at maintaining a constant supply of electricity, particularly when solar irradiance is low. The PV output peaked throughout the day, demonstrating notable diurnal fluctuation. The PV generation complemented the hydro generation, ranging from 0 kW at night to approximately 73 kW in the middle of the day. The sporadic grid import, which mostly occurs during hours 8–12 and 32–36 when load demand is high, suggests that it is used as a backup source to keep power balance when demand is high or renewable energy is insufficient.

Over 48 h, the load demand shows significant oscillations, ranging from 20 kW to over 286 kW. Every time, the hybrid design can meet the load, implying that the generating sources are dispatched in the best possible way. With a limited range of 93% to 100%, the battery's state of charge (SoC) was steadily high, suggesting little cycling activity. This stability means that the demand is mostly met by the combined hydro and PV power, which lessens the need for the energy balancing storage system. Additionally, the battery's near-saturation SoC levels imply that its primary role is that of a contingency reserve, preserving system dependability and absorbing excess generation during limited surplus times.

3.6 Comparison with similar studies

Comparing the current study to the related research may be challenging due to the complexity of system configurations, power component ratings, load demands, and RE sources at various locations. However, LCOE is a useful indicator for affirming the cost-effectiveness of hybrid energy systems (Ali et al., 2025a, b; Dodo et al., 2024; Hüner, 2025; Roy et al., 2024). Table 9 presents an overview of the LCOE of diverse hybrid energy systems in different parts of the world. As seen in the Table, the LCOE, which falls within the range

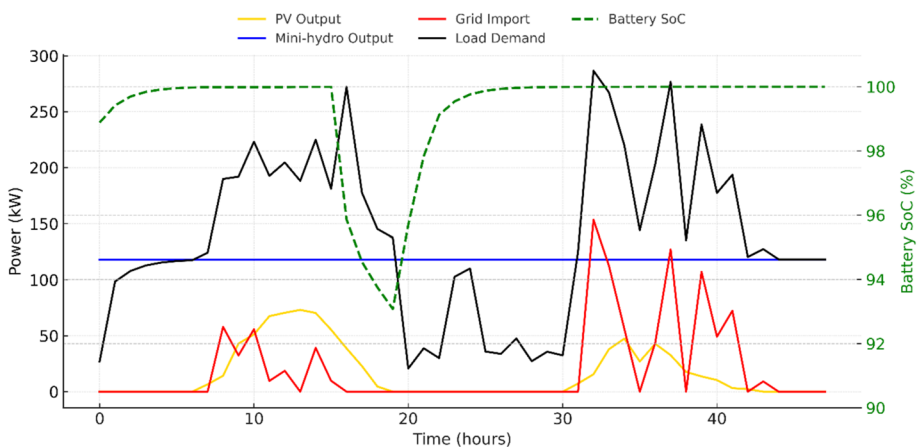


Fig. 11 Hourly dispatch and battery SoC profile

Table 9 Comparison with relevant literature

Region	HRES configuration	LCOE (\$/kWh)	Reference
Bangladesh	Grid/PV/WT	0.04	Ali et al., (2025a, b)
Morocco	PV/WT/FC/ Hydro	0.15	Allouhi (2024)
Indonesia	PV/Hydro/DG	0.31	Bawan et al. (2025)
India	Grid/WT/PV	0.15	Al Abri et al. (2023)
Jordan	Grid/PV/WT/DG	0.04	Al-Ra-washdeh et al. (2023)
Libya	PV/WT/FC	\$1.86	Yahya et al. (2024)
Turkey	Grid/PV/ Electrolyzer	0.21	Hüner (2025)
Algeria	PV/WT/DG	0.25	Bacha et al. (2024)
Jordan	WT/DG	0.24	Al-Omari et al. (2025)
India	PV/WT/DG	0.32	Roy et al. (2024)
Nigeria	Grid/PV/Hydro	0.13	This study

WT wind turbines, *PV* photovoltaic, *BG* biogas generator, *FC* fuel cells, *DG* diesel generator

of \$0.04 kWh to \$0.32/kWh, differs not only across the globe but even within the same region. This is because it relies upon the characteristics and economics of the components that constitute the energy systems and the resources available at the project sites. Similarly, the grid-connected systems have lower LCOE compared to the off-grid systems, implying that where there is grid reticulation, a grid-connected energy system is a better alternative than an off-grid energy system. Furthermore, the two prominent studies that also integrated hydro resources are Allouhi (2024) and Dodo et al. (2021). The relatively lower LCOE of Allouhi (2024) can be attributed to the exclusion of diesel generators from the system. A more economical system can be further achieved through grid integration. Therefore, the present study has a competitive LCOE compared to other studies in the literature, inferring the significance of exploring grid-connected hybrid hydro-solar systems for reliable, economical, and environmentally friendly energy solutions.

4 Conclusion

This study performed techno-economic assessments of four different energy systems: Grid/PV/Hydro, PV/Hydro, Grid/Hydro, and Grid/PV systems. In addition to using the community's run-of-river, HOMER's analysis took into account the constant utility grid outages to reflect the realities of developing nations. According to the results, the hybrid Grid/PV/hydro system is the best option for sustainably supplying the community's daily electricity demand of 2,822.20 kWh at the lowest NPC of \$9,160,308.00 and LCOE of \$0.13/kWh. With a renewable fraction of 92.10%, and emission characteristics which includes CO₂ of

34,460 kg/yr, CO of 2.73 kg/yr, PM of 12.80 kg/yr, SO₂ of 173 kg/yr, NO of 37.40 kg/yr, and UH of 0 kg/yr, the optimal system significantly minimizes reliance on fossil resources for electricity supply, and offers substantial emissions savings while maintaining cost-effectiveness. Despite having low operating costs and offering auxiliary grid services like frequency regulation, inertia, and black start, the run-of-river hydro component provides the majority of power (80.9%), but its performance is still influenced by seasonal river flows. Although solar PV makes up 12.4% of the generation and is resilient during dry seasons, its unpredictability may require additional investment in storage or innovative dispatching techniques. While reliability is guaranteed by the minimal grid share (6.8%), the system is vulnerable to grid instability threats. These dynamics highlight how crucial it is to strike a balance between PV sizing, battery integration, and seasonal hydro availability to ensure cost stability and year-round supply security.

By utilizing locally accessible renewable resources with little environmental impact, the proposed hybrid model provides a replicable route for rural and semi-urban electrification. The study offers a reliable path to tackling energy poverty and emissions reduction in developing economies by fusing solar energy's carbon-free generation with hydropower's stabilising benefits. Furthermore, integration into regional grid-planning frameworks can improve scalability. However, pilot-scale adoption is advised to confirm technical performance and community acceptance. Along with policy tools that can facilitate funding and long-term sustainability, future research should examine complementarities with waste-to-energy or agro-residue-based generation.

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Declarations

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References

- Adamu, A., & Dodo, U. A. (2025). Techno-economic evaluation of grid-connected hybrid energy system based on run-of-river and solar energy plants for sustainable electrification of a rural community. *ABUAD Journal of Engineering Research and Development (AJERD)*, 8(2), 1–13. <https://doi.org/10.53982/ajerd>
- Adetoro, S. A., Olatomiwa, L., Tsado, J., & Dauda, S. M. (2023). Techno-economic and environmental impact assessment of a hybrid renewable energy system employing an enhanced combined dispatch strategy. *Green Energy and Resources*, 1(4), 1–15. <https://doi.org/10.1016/j.gerr.2023.100044>

- Agoundedemba, M., Kim, C. K., Kim, H. G., Nyenge, R., & Musila, N. (2025). Modelling and optimization of microgrid with combined genetic algorithm and model predictive control of PV/Wind/FC/battery energy systems. *Energy Reports*, 13, 238–255. <https://doi.org/10.1016/j.egy.2024.12.008>
- Akhator, P. E., Obanor, A. I., & Sadjere, E. G. (2019). Electricity situation and potential development in Nigeria using off-grid green energy solutions. *Journal of Applied Sciences and Environmental Management*, 23(3), 527–537. <https://doi.org/10.4314/jasem.v23i3.24>
- Akinte, O. O., Plangklang, B., Prasartkaew, B., & Aina, T. S. (2023). Energy storage management of a solar photovoltaic-biomass hybrid power system. *Energies*, 16(13), 1–32. <https://doi.org/10.3390/en16135122>
- Al Abri, A., Al Kaaf, A., Allouyahi, M., Al Wahaibi, A., Ahshan, R., Al Abri, R. S., & Al Abri, A. (2023). Techno-economic and environmental analysis of renewable mix hybrid energy system for sustainable electrification of Al-Dhafrat rural area in Oman. *Energies*, 16(1), 1–15. <https://doi.org/10.3390/en16010288>
- Ali, F., Halim, M. A., Julhash, M. M., & Ashikuzzaman, M. (2025a). Economic and environmental benefits of grid-connected PV-biomass systems in a Bangladeshi University: A HOMER pro approach. *International Transactions of Electrical Energy Systems*, 2025, 1–24. <https://doi.org/10.1155/etep/5053853>
- Ali, M. F., Sheikh, M. R. I., Akter, R., Islam, K. M. N., & Ferdous, A. H. M. I. (2025b). Grid-connected hybrid microgrids with PV/wind/battery: Sustainable energy solutions for rural education in Bangladesh. *Results in Engineering*, 25, 1–21. <https://doi.org/10.1016/j.rineng.2024.103774>
- Allouhi, A. (2024). A hybrid PV/wind/battery energy system to assist a run-of-river micro-hydropower for clean electrification and fuelling hydrogen mobility for young population in a rural Moroccan site. *Journal of Cleaner Production*, 442, 1–16. <https://doi.org/10.1016/j.jclepro.2024.140852>
- Al-Omari, Z., Khlaifat, N., & Haddad, M. (2025). A feasibility study of combining solar/wind energy to power a water pumping system in Jordan's desert/Al-Mudawwara village. *Environmental and Sustainability Indicators*, 25, 1–11. <https://doi.org/10.1016/j.indic.2024.100555>
- Al-Quraan, A., & Al-Mhairat, B. (2024). Sizing and energy management of grid-connected hybrid renewable energy systems based on techno-economic predictive technique. *Renewable Energy*, 228, 120639. <https://doi.org/10.1016/j.renene.2024.120639>
- Al-Rawashdeh, H., Al-Khashman, O. A., Al Bdour, J. T., Gomaa, M. R., Rezk, H., Marashli, A., et al. (2023). Performance analysis of a hybrid renewable-energy system for green buildings to improve efficiency and reduce GHG emissions with multiple scenarios. *Sustainability (Basel)*, 15(9), 1–32. <https://doi.org/10.3390/su15097529>
- Bacha, B., Ghodbane, H., Dahmani, H., Betka, A., Toumi, A., & Chouder, A. (2024). Optimal sizing of a hybrid microgrid system using solar, wind, diesel, and battery energy storage to alleviate energy poverty in a rural area of Biskra, Algeria. *Journal of Energy Storage*, 84, 1–19.
- Bawan, E. K., Wijaya, F. D., Ali, H. R., Member, S., Vasquez, J. C., & Member, S. (2025). Comprehensive analysis and optimal design of hybrid renewable energy system for rural electrification in West Papua, Indonesia. *IEEE Access*, 13, 1–26.
- Bayazit, Y. (2021). The effect of hydroelectric power plants on the carbon emission: An example of Gökçekaya dam, Turkey. *Renewable Energy*, 170, 181–187. <https://doi.org/10.1016/j.renene.2021.01.130>
- Demir, H. (2024). Design and optimization of hybrid renewable energy systems for hydrogen production at Aksaray University campus. *Process Safety and Environmental Protection*, 192, 543–556. <https://doi.org/10.1016/j.psep.2024.10.080>
- Dodo, U. A., & Ashigwuike, E. C. (2023). In-depth physico-chemical characterisation and estimation of the grid power potential of municipal solid wastes in Abuja city. *Energy Nexus*, 10, 1–9. <https://doi.org/10.1016/j.nexus.2023.100192>
- Dodo, U. A., Ashigwuike, E. C., & Eronu, E. M. (2021). Renewable energy readiness in Nigeria: A review focusing on power generation. *Uniabuja Journal of Engineering and Technology*, 1(1), 115–144.
- Dodo, U. A., Ashigwuike, E. C., & Emechebe, J. N. (2022). Techno-economic evaluation of municipal solid waste-fueled biogas generator as a backup in a decentralized hybrid power system. *Process Integration and Optimization for Sustainability*. <https://doi.org/10.1007/s41660-022-00223-9>
- Dodo, U. A., Salami, B. A., Bashir, F. M., Hamdoun, H. Y., Rashed Alsadun, I. S., Dodo, Y. A., et al. (2024). Investigating the influence of erratic grid on stationary battery energy storage technologies in hybrid power systems: Techno-enviro-economic perspectives. *Energy*, 304, 1–16. <https://doi.org/10.1016/j.energy.2024.132140>
- Elmasides, C., Kosmadakis, I. E., & Athanasiou, C. (2025). A comprehensive power management strategy for the effective sizing of a PV hybrid renewable energy system with battery and H2 storage. *Journal of Energy Storage*, 106, 1–25.
- Ezekwem, C., Muthusamy, S., & Ezekwem, P. C. (2024). Optimal selection and design of grid-connected hybrid renewable energy system in three selected communities of Rivers State. *Scientific African*, 25, 1–31. <https://doi.org/10.1016/j.sciaf.2024.e02305>

- Frate, G. F., Ferrari, L., & Desideri, U. (2021). Energy storage for grid-scale applications: Technology review and economic feasibility analysis. *Renewable Energy*, 163, 1754–1772. <https://doi.org/10.1016/j.renene.2020.10.070>
- He, Y., Guo, S., Dong, P., Zhang, Y., Huang, J., & Zhou, J. (2023). A state-of-the-art review and bibliometric analysis on the sizing optimization of off-grid hybrid renewable energy systems. *Renewable and Sustainable Energy Reviews*, 183(1–26). <https://doi.org/10.1016/j.rser.2023.113476>
- Hüner, B. (2025). Feasibility and environmental analysis of biogas-based hybrid energy system using HOMER pro software: A case study for Hatay. *Energy Conversion and Management*, 326, 1–15.
- Husein, M., Moner-Girona, M., Falchetta, G., Stevanato, N., Fahl, F., & Szabó, S. (2024). The impacts of incentive policies on improving private investment for rural electrification in Nigeria – A geospatial study. *Heliyon*. <https://doi.org/10.1016/j.heliyon.2024.e27440>
- Hussain Alhamami, A., Abba, S. I., Musa, B., Aminu Dodo, Y., Abiodun Salami, B., Dodo, U. A., & Alyami, S. H. (2024). Off-grid multi-region energy system design based on energy load demand estimation using hybrid nature-inspired optimization algorithms. *Energy Conversion and Management*, 315, 1–17. <https://doi.org/10.1016/j.enconman.2024.118766>
- Ibrahim, M., Imam, Y., & Ghanem, A. (2019). Optimal planning and design of run-of-river hydroelectric power projects. *Renewable Energy*, 141, 858–873. <https://doi.org/10.1016/j.renene.2019.04.009>
- Jumare, I. A., Bhandari, R., & Zerga, A. (2020). Assessment of a decentralized grid-connected photovoltaic (PV)/wind/biogas hybrid power system in northern. *Energy, Sustainability and Society*, 10(30), 1–25.
- Malik, P., Awasthi, M., & Sinha, S. (2022). A techno-economic investigation of grid integrated hybrid renewable energy systems. *Sustainable Energy Technologies and Assessments*, 51, 1–15. <https://doi.org/10.1016/j.seta.2022.101976>
- Nemouchi, W., Amrane, Y., Nemouchi, H., & Boucetta, N. L. (2024). Energy management for optimal design of PV/wind/diesel system for water pumping irrigation in semi-arid climate. *Energy Conversion and Management*, 304, 1–14. <https://doi.org/10.1016/j.enconman.2024.118216>
- Odetoeye, O. A., Olulope, P. K., Olanrewaju, O. M., Alimi, A. O., & Igbinosa, O. G. (2023). Multi-year techno-economic assessment of proposed zero-emission hybrid community microgrid in Nigeria using HOMER. *Heliyon*, 9(9), 1–21. <https://doi.org/10.1016/j.heliyon.2023.e19189>
- Oladigbolu, J. O., Ramli, M. A. M., & Al-Turki, Y. A. (2020). Optimal design of a hybrid PV solar/micro-hydro/diesel/battery energy system for a remote rural village under tropical. *Electronics*, 9(1491), 1–20.
- Ondachi, P. (2024). Optimisation of an on-grid hybrid energy system: A case study of the main campus of the university of Abuja, Nigeria. *ABUAD Journal of Engineering Research and Development (AJERD)*, 7(1), 295–308. <https://doi.org/10.53982/ajerd>
- Ozigis, I. I., Oodo, S. O., & Muhammad, I. D. (2020). A feasibility study of mini-hydroelectric power plant for seasonal base load. *Arab Zone Journal of Engineering, Technology & Environment*, 15, 124–145. <https://azojete.com.ng/index.php/Azojete/article/view/20/0>. Accessed 20 Mar 2025.
- Rajbongshi, R., & Rajkumari, A. (2025). Optimization and economic analysis of hybrid renewable energy system. *Energy*, 317, 1–8.
- Riayatsyah, T. M. I., Geumpana, T. A., Rizwanul Fattah, I. M., Rizal, S., & Indra Mahlia, T. M. (2022). Techno-economic analysis and optimisation of campus grid-connected hybrid renewable energy system using HOMER Grid. *Sustainability (Basel)*, 14(13), 1–18. <https://doi.org/10.3390/su14137735>
- Roy, D., Wang, R., Roy, S., Smallbone, A., & Paul, A. (2024). Hybrid renewable energy systems for sustainable power supply in remote location: Techno-economic and environmental assessment. *Energy Conversion and Management: X*, 24, 1–17. <https://doi.org/10.1016/j.ecmx.2024.100793>
- Saddari, N., Agyemang Derkyi, N. S., Peprah, F., Gyamfi, S., & Donkor, G. K. (2025). Techno-economic and environmental assessment of grid and solar photovoltaic microgrid supply options for isolated off-grid rural communities toward sustainable and affordable electricity in Nkoranza South, Bono East, Ghana. *Results in Engineering*. <https://doi.org/10.1016/j.rineng.2025.103915>
- Salisu, S., Mustafa, M. W., Olatomiwa, L., & Mohammed, O. O. (2019). Assessment of technical and economic feasibility for a hybrid PV-wind-diesel-battery energy system in a remote community of north central Nigeria. *Alexandria Engineering Journal*, 58(4), 1103–1118. <https://doi.org/10.1016/j.aej.2019.09.013>
- Shen, W., Qiu, J., & Dong, Z. (2018). Electricity network planning targeting low-carbon energy transition. *Global Energy Interconnection*, 1(4), 487–499. <https://doi.org/10.14171/j.2096-5117.gei.2018.04.009>
- Syahputra, R., & Soesanti, I. (2021). Renewable energy systems based on micro-hydro and solar photovoltaic for rural areas: A case study in Yogyakarta, Indonesia. *Energy Reports*, 7, 472–490.
- Vu, B. H., Husein, M., Kang, H. K., & Chung, I. Y. (2019). Optimal design for a campus microgrid considering ESS discharging incentive and financial feasibility. *Journal of Electrical Engineering & Technology*, 14(3), 1095–1107. <https://doi.org/10.1007/s42835-019-00142-9>

- Yadav, S., Kumar, P., & Kumar, A. (2024). Techno-economic assessment of hybrid renewable energy system with multi energy storage system using HOMER. *Energy*, 297, 1–21. <https://doi.org/10.1016/j.energy.2024.131231>
- Yahya, W., Saied, K. M., Nassar, A., Qader, M. R., Al-Nehari, M., Zarabia, J., & Jian, Z. (2024). Optimization of a hybrid renewable energy system consisting of a of PV/wind turbine/battery/fuel cell integration and component design. *International Journal of Hydrogen Energy*, 94, 1406–1418. <https://doi.org/10.1016/j.ijhydene.2024.11.187>

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